MRI Reconstruction

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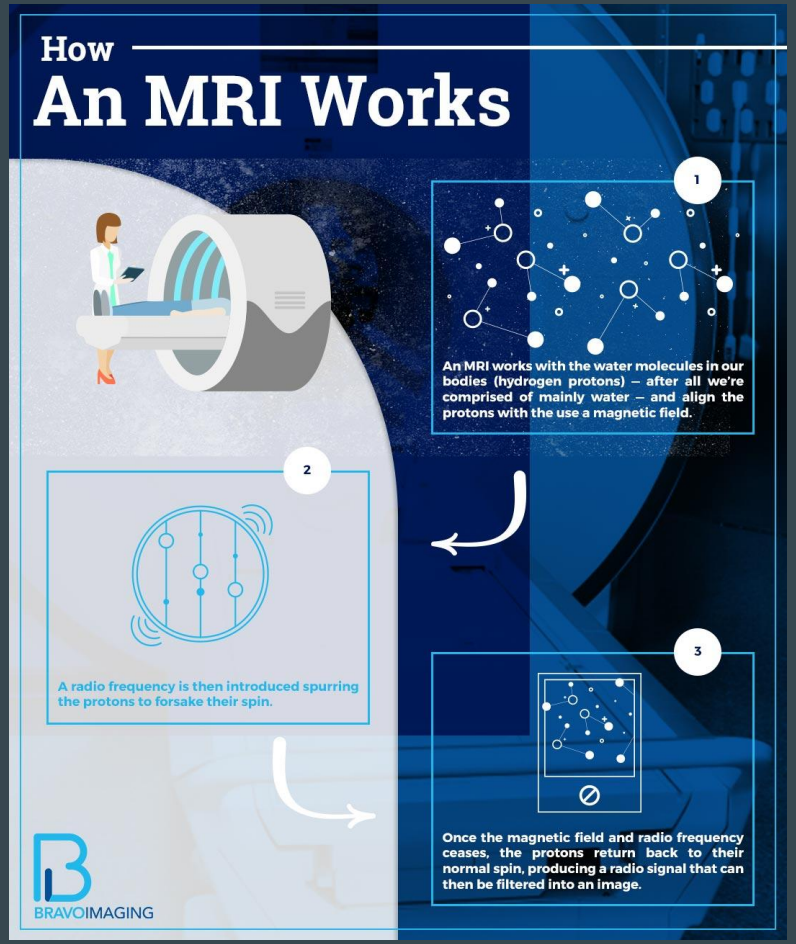
Background

MRI machines don't actually take pictures: they use magnets and the physical properties of water in our bodies to capture radio-frequency data.

Each radio signal is converted to a pixel in an image using a Fourier transform. By convention, we say that the MRI stores radio-frequency data in *k-space*, before being converted to an image.



Source: Bravo Imaging

The Problem

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Staying still for a long time is hard for

- Children
- Dementia Patients
- Patients with Huntington's, addictions, etc.

Shorter scan times = more accessible treatment

Potential Solution

What if, instead of measuring every value in k-space, we only directly measured a subset of them?

- Quicker scans, but more blurry images

Can we can **reconstruct** a partially formed MRI image using optimization?

- Quick scans with more detailed images

Mathematics

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

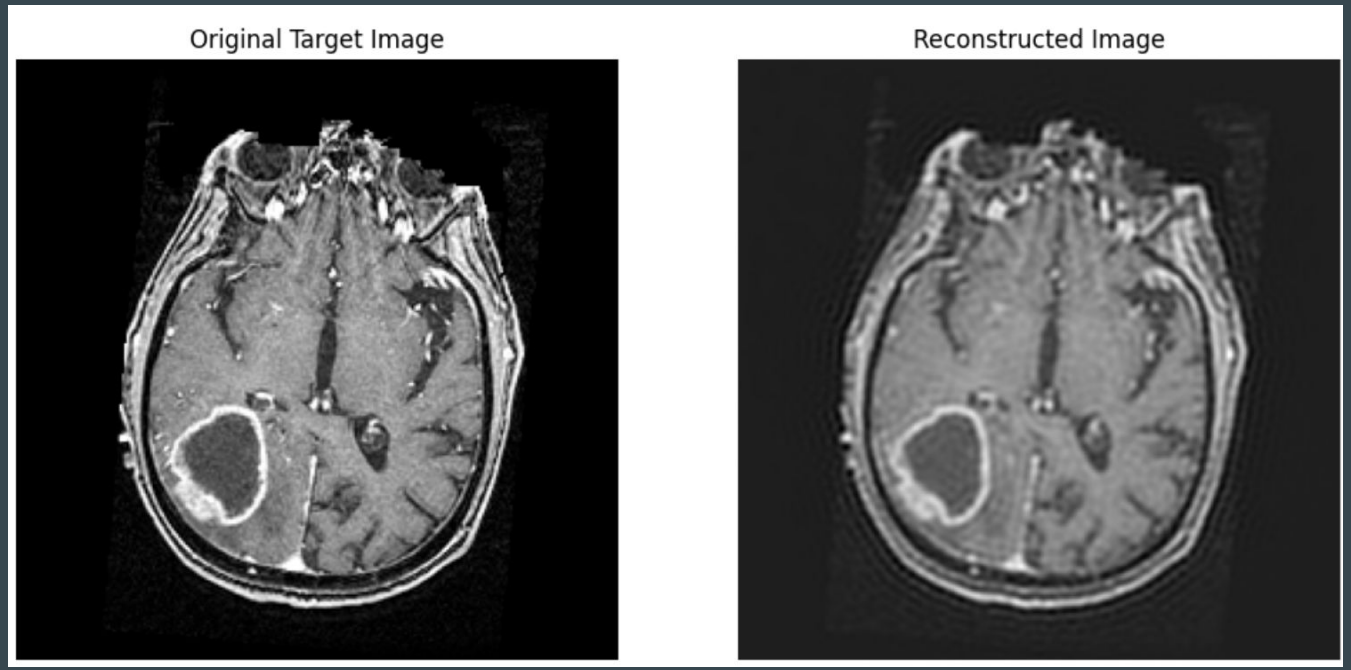
$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We use **Gaussian kernels** to impose smoothness.

Implementation

- **Data** : UPenn-GBM Cancer Imaging Archive. Selected target image.
  - Create kernel basis & lots of data preparation
- **Simulate** the under-sampling of frequencies: 'masking' the data by adding a filter to only show low-frequency waves.
- **Solve** : Using an Adam (Adaptive Moment Estimation) optimizer.
- **Validate** : (1) compare reconstructed image to original MRI target image and (2) measure loss over iterations.

Results



Pro:

- Working prototype w/real data

Con:

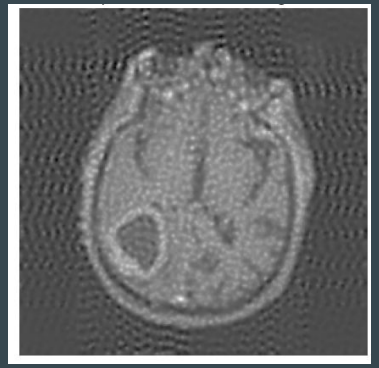
- Blurry image
- No frame of reference for accuracy

Comparing Kernels vs. Baseline Reconstruction

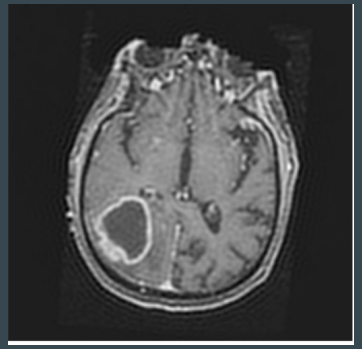
Metric	*LaPlacian Kernels*	*Gaussian Kernels*	*Zero-Filled FFT*
MSE	0.0186	0.00529	0.010049
PSNR	17.30 dB	22.78 dB	19.98 dB

Original Image

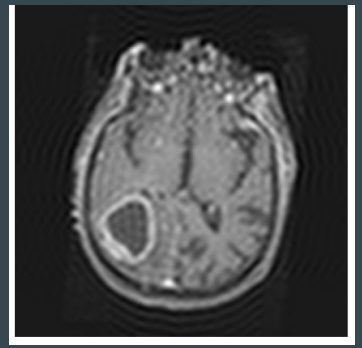
Laplace Reconstruction



Gaussian Reconstruction



Zero-filled FFT Reconstruction

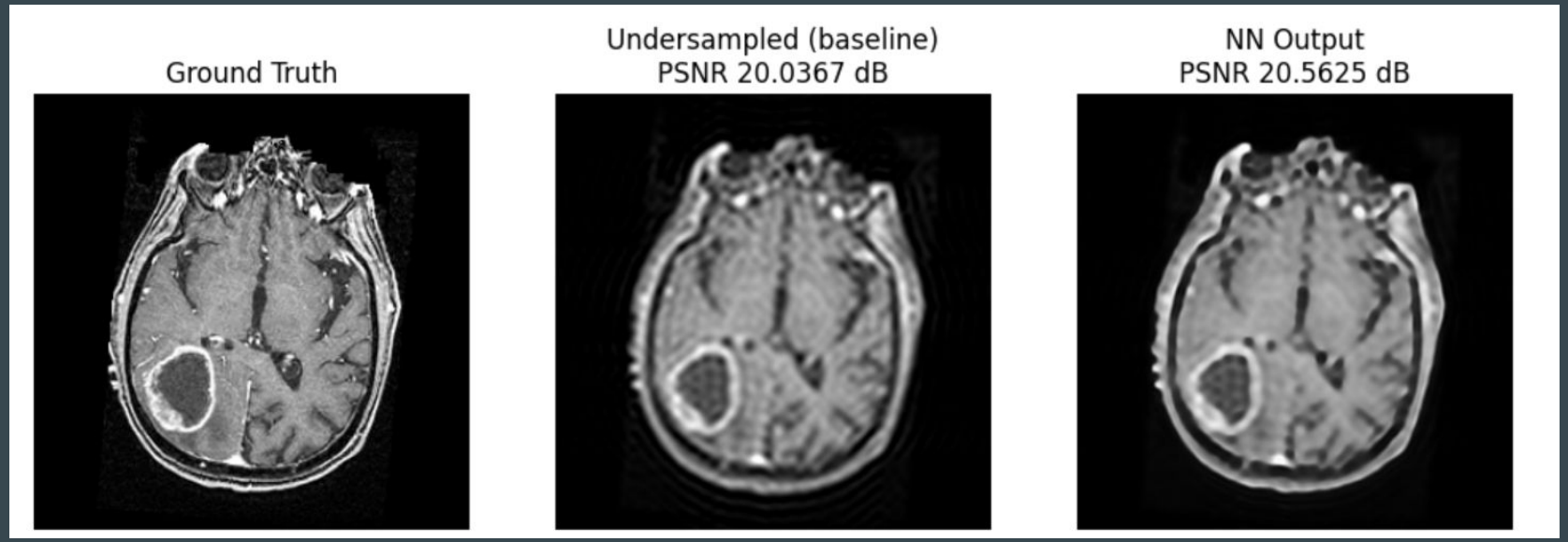


Gaussian kernel reconstruction provides highest clarity.

Another Baseline: The Traditional Approach

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small CNN:
  - 3 layers, 64 channels, 3x3 kernels (standard, small)
  - Input: undersampled MRI slice
  - Predicts a correction: $x* = x + f(x)$ where
    - x^* is final reconstructed image
    - $f(x)$ is learned correction
    - x is the undersampled input
  - Used L1 loss (common in image restoration tasks): $||x + f(x) - x_{true}||$

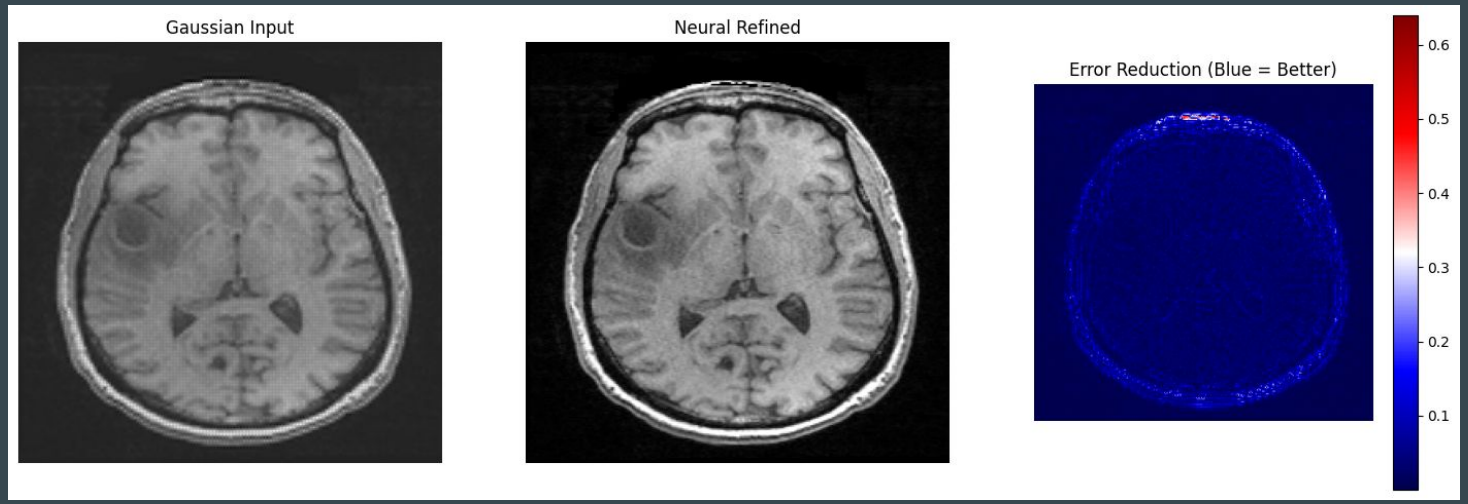
Neural Network Baseline Results



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

Warm-Start Neural Network

Again, used kernel reconstruction as input to a neural network but only let the neural network fill in the directions the kernel did not already smooth. We also changed the neural network architecture to U-Net, rather than ReLU which is more common in medical applications.



Warm-Start Neural Network Results

	Gaussian Reconstruction	Warm-Start NN Reconstruction
PSNR	19.581 dB	137.75 dB
MSE	0.011	0
SSIM	0.605	1.0

Achieved a perfect reconstruction of the desired image!

Super Resolution Convolutional Neural Network (SRCNN)

SRCNN is a three-layer CNN for single-image super resolution. Instead of applying kernels, we used SRCNN to reconstruct the entire slice. It made the following improvements:

	Kernel Optimizer (Gaussian)	SRCNN	Improvement
PSNR	28.78 dB	37.74 dB	8.96 dB
SSIM	0.8786	0.9534	0.0748

Next steps

- **Train / test across multiple slices:** so far, we have mostly been testing on a single slice to show proof of concept. Now we need to test across multiple slices
- **Diffusion models:** see if replacing the warm-start neural network with diffusion model improves reconstruction since diffusion models used in image generation
- **Multi-image super resolution:** add context by considering the slices before/after the slice that is being reconstructed
- **Images & Videos** : try our reconstruction approaches on various images and videos

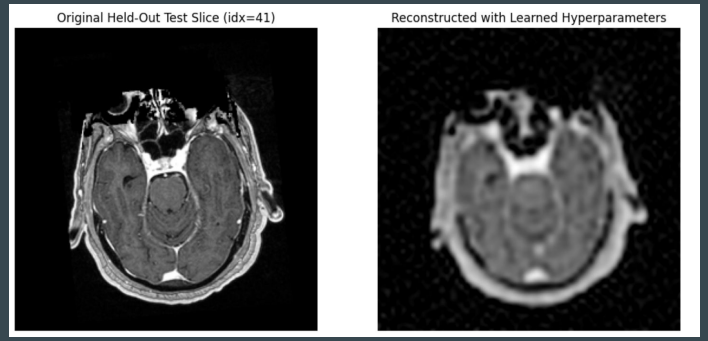
Multi-Slice Training: Setup

- **Adding a TV penalty:** Our optimization equation largely remained the same, with the addition of a TV regularizer:
- **Parameter Tuning**
  - Training for kernel width, number of kernels, and TV penalty.
  - Using both Gaussian and LaPlacian kernels to align with literature.
- **Training Steps** :
  - Choose candidate hyperparameters,
  - Reconstruct a set of training slices with those hyperparameters
  - Compute reconstruction metrics for each slice.
  - Average metrics across the slices and selected the hyperparameters that performed the best on the test slices.

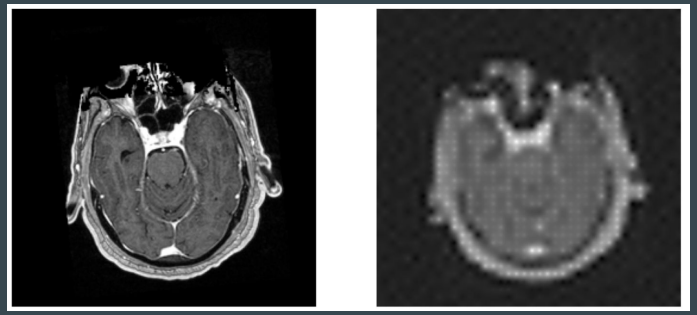
Multi-Slice Training

Kernel Type	Sigma	Number of Kernels	TV Weight	Test MSE	Test PSNR
LaPlacian	4.1017	1600	1.3145e-03	0.009615	20.173 dB
Gaussian	2.5	3844	0.0	0.007886	21.032 dB

LaPlacian



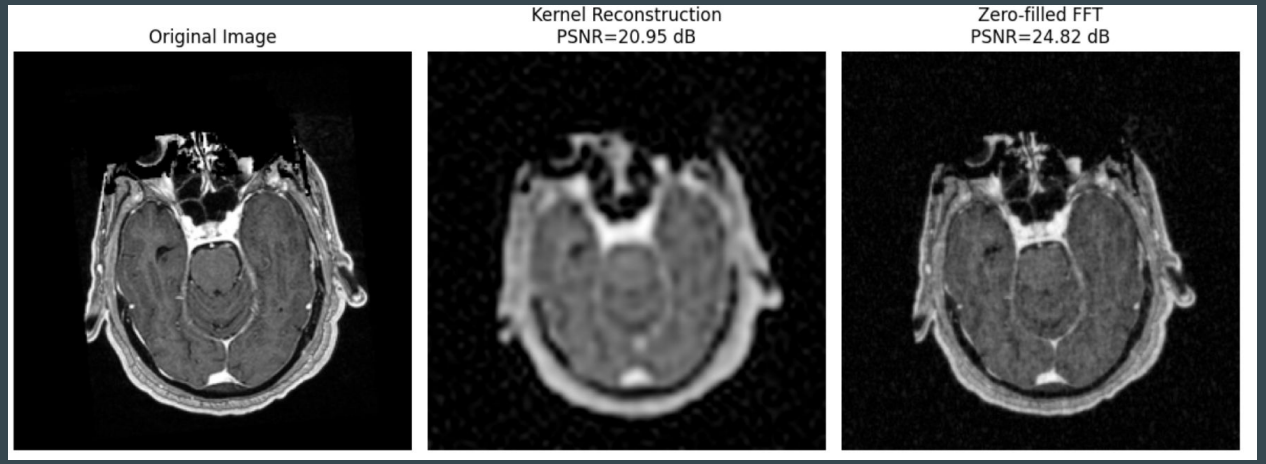
Gaussian



Like in the single-slice model, the Gaussian kernels performed better as measured by MSE and PSNR.

Multi-Slice Training: Zero-Filled FFT

- Lower PSNR for multi-slice zero-filled FFT compared to the single-slice model.
- Simpler method with lower computational time makes zero-filled a good candidate for more advanced models.



Residual Convolutional Neural Networks (CNN) - Setup

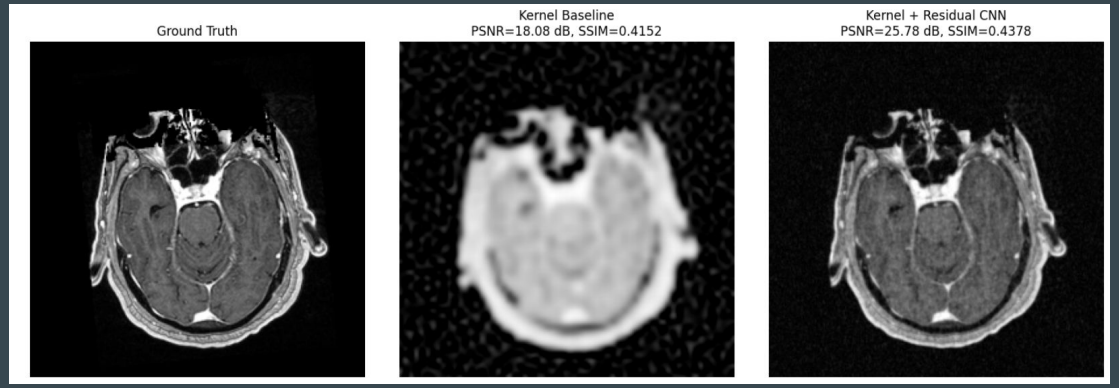
Basic setup: $x_{output} = x_{input} + f(x_{input})$

- E.g. CNN output = Warm-start (from zero-filled FFT) + learned residual correction
- Restriction: Model cannot change measured k-space values
- Tested with zero-filled FFT, Gaussian kernels, and LaPlacian kernels.

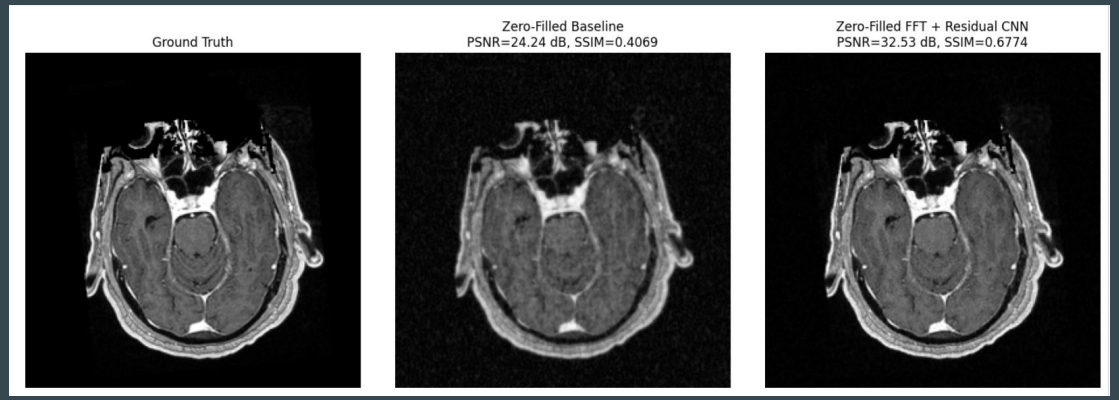
	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Residual CNN	0.002479	26.06 dB	0.4153
Zero-filled FFT Reconstruction & Residual CNN	0.000478	33.20 dB	0.6758

Residual CNN - Results

Kernel Reconstructions



Zero-Filled FFT Reconstructions



- Residual CNN consistently shows better performance than kernel reconstruction alone
- Zero-filled FFT outperforms kernel reconstruction on PSNR, MSE, SSIM, and compute time.

Diffusion Models

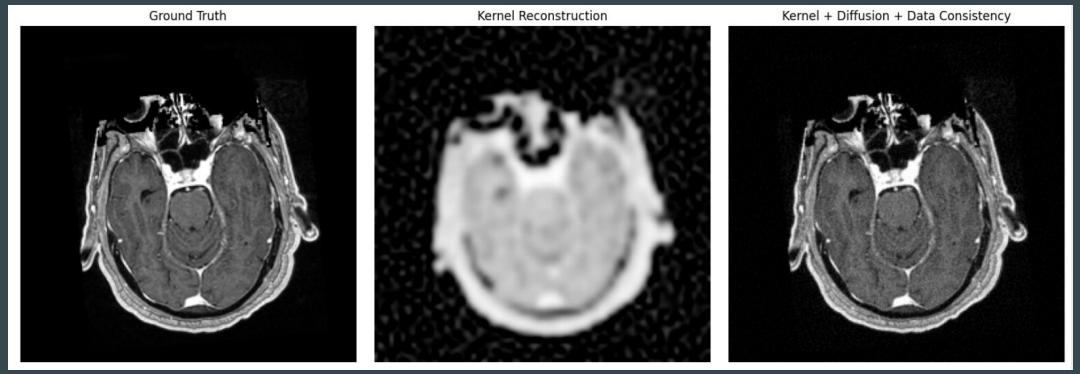
Diffusion models are commonly used in image reconstruction by iteratively adding noise to training images and producing image output from gradually removing noise layers from a pure-noise image to recover the original image.

- Step 1: Train diffusion model on the initial reconstruction (kernel or zero-filled FFT)
- Step 2: Model iteratively corrects the areas of the image that were not measured k-space data (i.e., “real” parts of the image).

	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Diffusion Model Correction	0.000690	31.61 dB	0.7796
Zero-filled FFT Reconstruction & Diffusion Model Correction	0.000772	31.12 dB	0.7620

Diffusion Model: Results

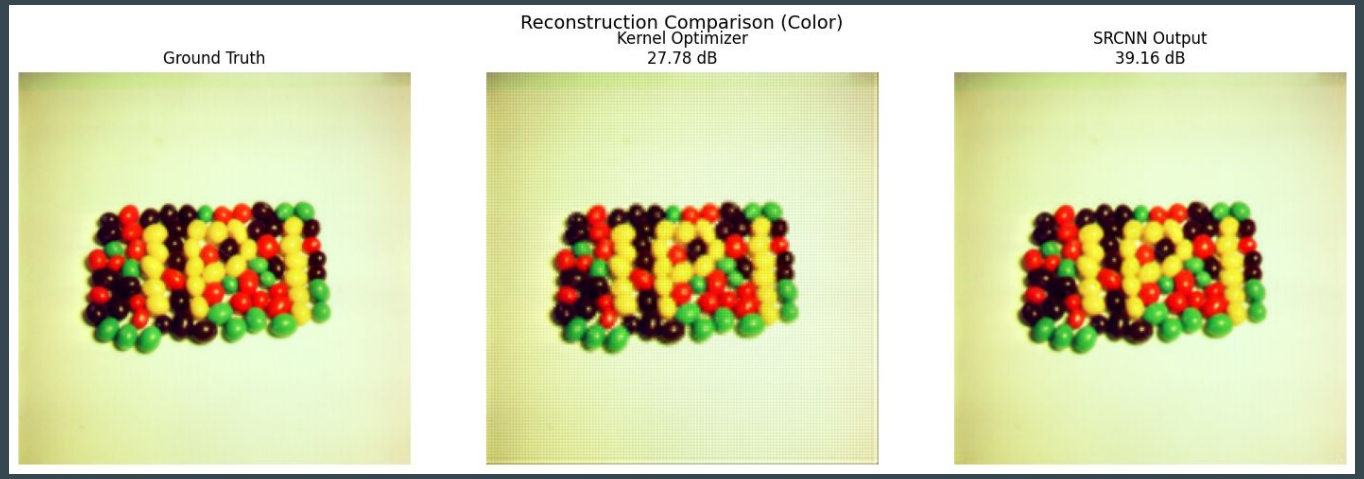
Kernel Diffusion Model



Zero-Filled FFT Diffusion Model

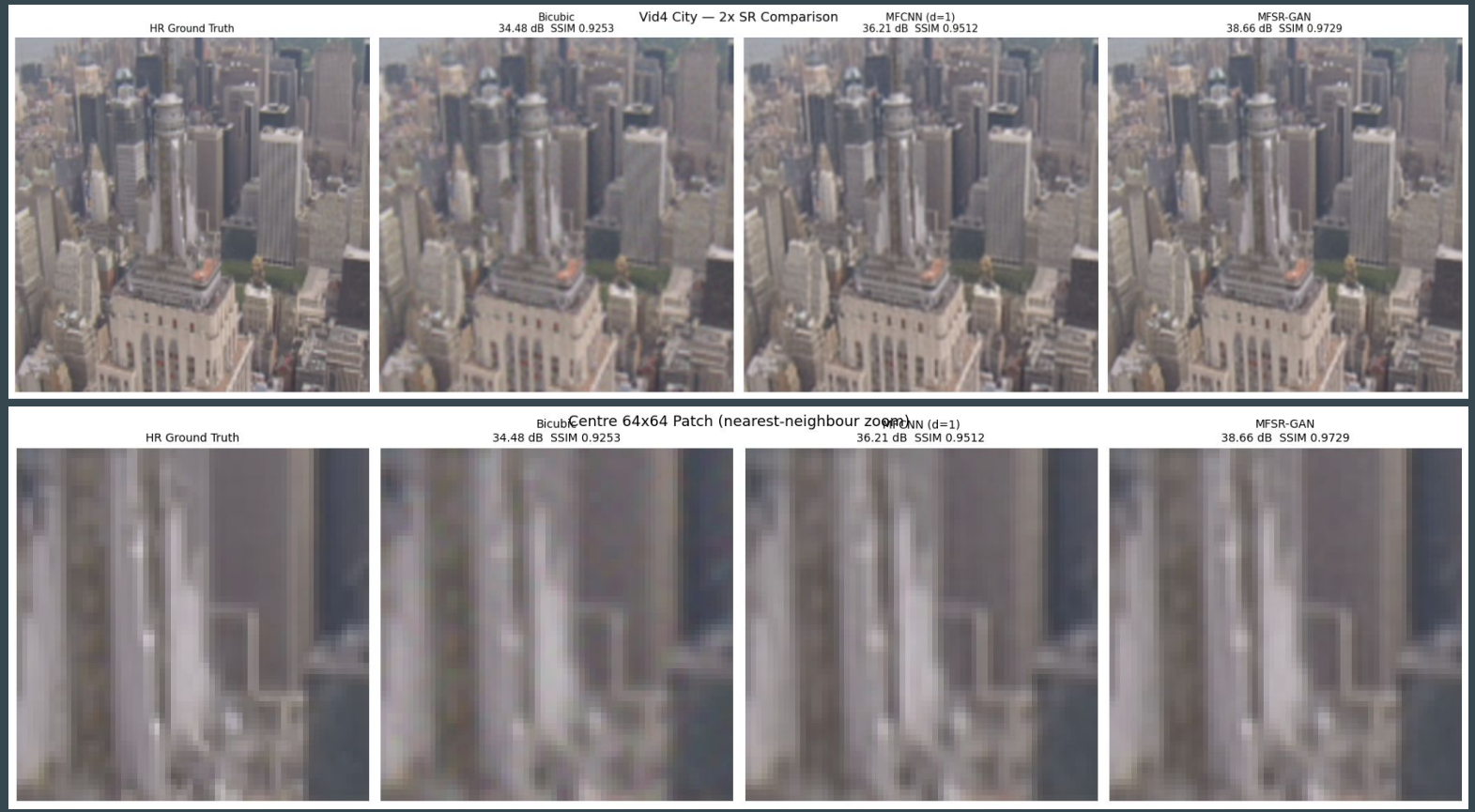
MSE was lower and PSNR and SSIM were higher for the kernel reconstruction compared to the zero-filled FFT, showing that the **combination of the kernel reconstruction and diffusion model actually does better than the baseline approach.**

Colored SRCNN



Updated the SRCNN to work with color and currently testing on deeper levels of SRCNN and training on multiple images

Multiframe CNN Methods



Multiframe CNN Methods(Continued)

We have implemented two types of Multiframe CNN methods which take into account the context of neighboring frames to help strengthen the reconstruction of each frame

MFCNN (Greaves & Winter, 2016)

A lightweight multi-frame CNN approach. All 5 LR burst frames are bicubic-upsampled to HR size, then concatenated along the channel axis (giving 15 channels for a d=1 window). A simple 5-layer conv net maps this to the SR output.

MFSR-GAN (Khan et al., CVPRW 2025)

A more sophisticated GAN-based approach with:

- Reference Difference Computation, deformable convolutions for motion-aware alignment, channel attention for multi-frame features, RRDB (Residual-in-Residual Dense Block), trained with a relativistic GAN loss + L1 loss

This initial test was done on a standard Vid4 “city” benchmark where each frame was downsampled and had noise added to simulate a blurry image

The next steps are to apply these methods to MRI imaging to see if they improve reconstruction as well as training the model on more data

Next Steps

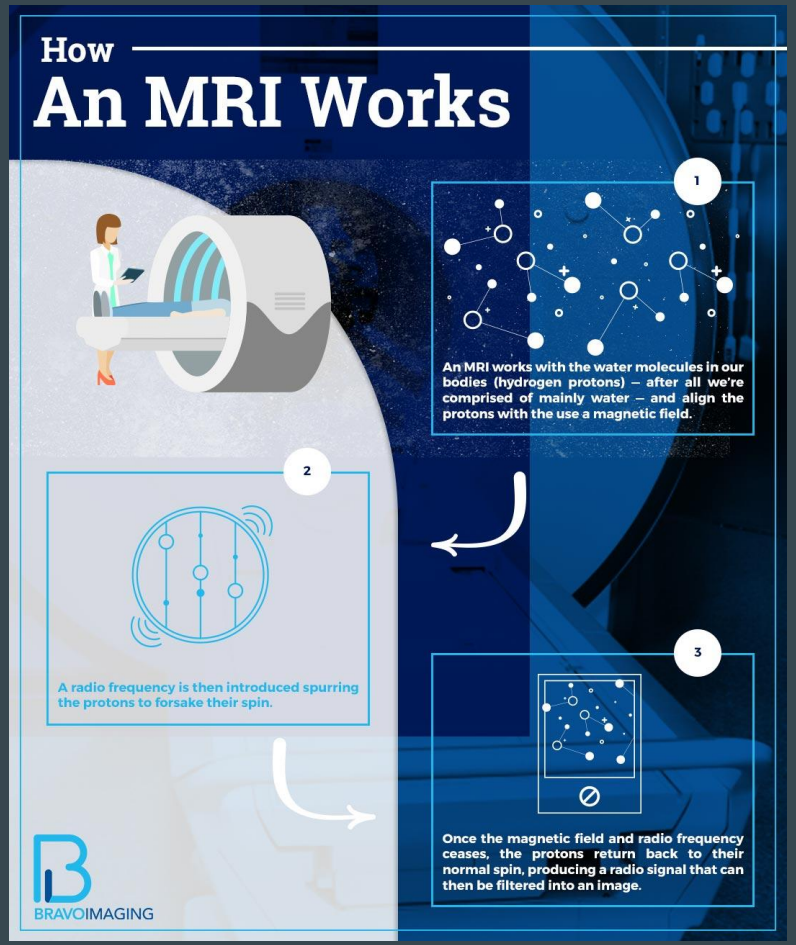
- Self-tuning diffusion models: to further reduce noise floor
- Multi-frame super resolution models: allows model to use context from neighboring images in the reconstruction
- Extension to non-medical imaging: Providing better baseline
- Improving zero-filled FFT instead of the kernel reconstruction
- Training SRCNN and MFCNN on more data

Lightning Presentations: MRI Reconstruction

Problem Statement & Motivation

There are a lot of pixels in a typical MRI image, which means the machine has to send out a lot of radio-frequencies, which **takes a long time** . This is **expensive** for hospitals operating MRI machines and **uncomfortable** for patients, who have to stay still while images are taken.

Shorter scan times = more accessible treatment



Technical Approach

A = MF

- M = Diagonal Matrix (entries are 1 if frequency is being measured, 0 if not)
- F = Discrete Fourier Transform

Given observed undersampled k-space data *y_obs,* we are looking for α that solves

$$\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$$

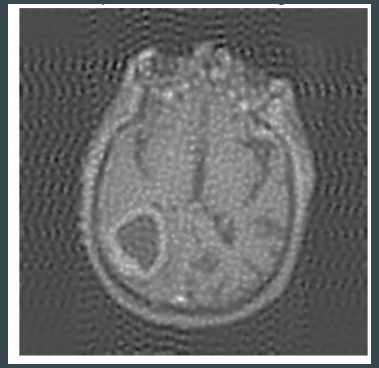
There are many possible solutions, so we restrict the reconstruction to a class of kernel basis functions for uniqueness. We start with **Gaussian and LaPlacian kernels** .

Initial Results pt. 1

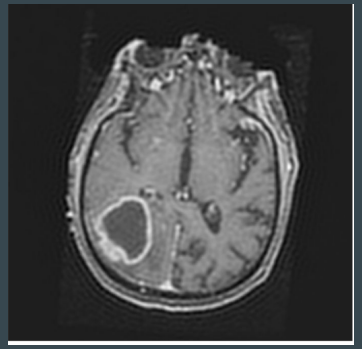
Metric	*LaPlacian Kernels*	*Gaussian Kernels*	*Zero-Filled FFT*
MSE	0.0186	0.00529	0.010049
PSNR	17.30 dB	22.78 dB	19.98 dB

Original Image

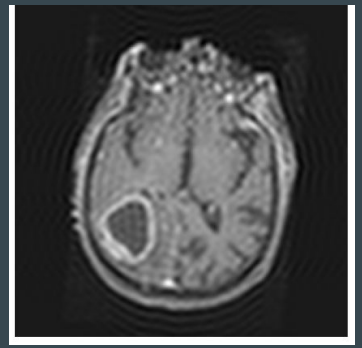
Laplace Reconstruction



Gaussian Reconstruction



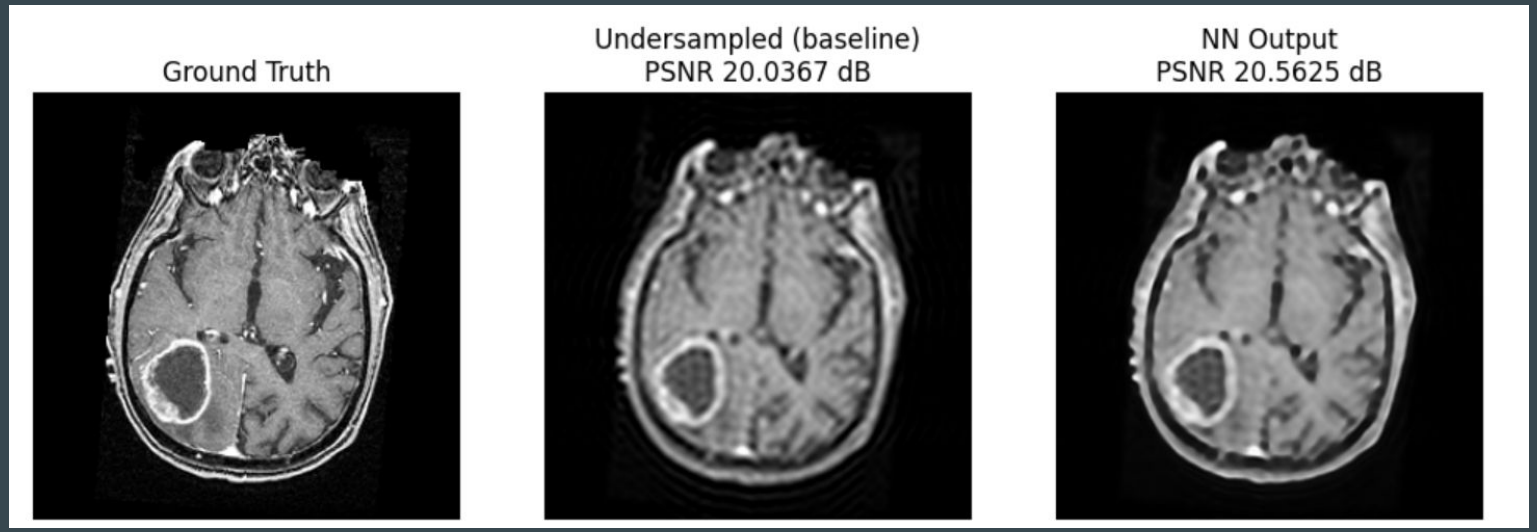
Zero-filled FFT Reconstruction



Gaussian kernel reconstruction provides highest clarity.

Initial Results pt. 2

- Modern MRI reconstruction often uses learned priors (CNNs/diffusion models)
- Trained a small residual CNN, then reconstructed the same slice we used the kernel method on



Gaussian kernel still achieves highest PSNR, implying the kernel method is competitive to traditional, learned methods.

Next Steps

- Moving toward more general image reconstruction/upscaling due to difficulty in determining medical benchmarks
- Combining the kernel approach with neural networks
  - Use kernels to get a smooth reconstruction
  - Feed the reconstruction into a neural network
  - Network learns the residual and works as a residual corrector
  - Can only modify the unobserved regions of k-space
- Using the Structural Similarity Index as another metric of success

$$SSIM(I_{LR}, I_{HR}) = \frac{(2\mu_{I_{LR}}\mu_{I_{HR}} + c_1)(2\sigma_{I_{LR},I_{HR}} + c_2)}{(\mu_{I_{LR}}^2 + \mu_{I_{HR}}^2 + c_1)(\sigma_{I_{LR}}^2 + \sigma_{I_{HR}}^2 + c_2)}$$

- Explore different deep learning and neural network upsampling strategies (Pre, Post, and Progressive)